SWAPS – MEANING

A **swap** is an agreement between two parties to exchange cash flows in the future according to predetermined rules.

The cash flows may be based on:

- Interest rates
- Currencies
- Commodity prices
- Equity prices
- Credit events

The most common swaps are:

1. **Interest Rate Swap**
2. **Currency Swap**
3. Commodity Swap
4. Equity Swap
5. Credit Default Swap

Simple meaning

Suppose Company A has borrowed money at a **floating interest rate**, but it is worried that interest rates may increase.

Company B has borrowed money at a **fixed interest rate**, but it expects interest rates to fall.

They may enter into a swap agreement to exchange their interest payment obligations.

Basic illustration

SWAP AGREEMENT

Company A <----------> Company B

Exchange

Cash Flows

Fixed Interest <----------> Floating Interest

The companies generally continue to make payments to their original lenders. The swap arrangement separately determines the payments exchanged between the two companies.

IMPORTANT FEATURES OF SWAPS

1. Private agreement

A swap is normally an agreement between two parties, often arranged through a financial institution.

2. Over-the-counter derivative

Most traditional swaps are **OTC (Over-the-Counter)** contracts rather than exchange-traded contracts.

3. Notional principal

Swaps normally use a **notional principal amount**.

The notional amount is used for calculating interest payments but is generally **not exchanged** in an ordinary interest rate swap.

4. Fixed period

The swap agreement specifies:

- Start date
- Maturity date
- Payment dates
- Notional amount
- Interest rate
- Currency
- Calculation method

5. Exchange of cash flows

The parties exchange agreed cash flows during the life of the contract.

WHY DO COMPANIES USE SWAPS?

Swaps are primarily used for:

A. Risk management

Companies can reduce exposure to:

- Interest-rate risk
- Currency risk
- Commodity-price risk
- Credit risk

B. Reduction of financing cost

A company may be able to borrow more cheaply in one market and then use a swap to obtain its preferred type of financing.

C. Speculation

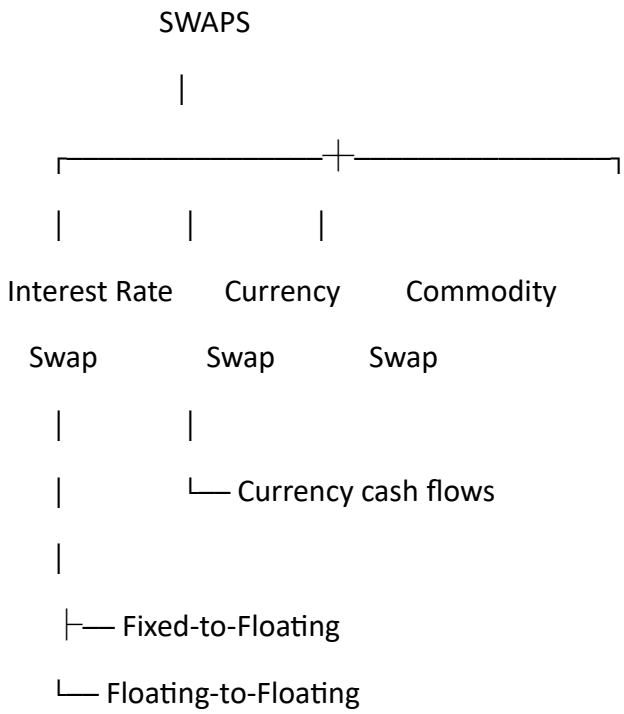
A company or financial institution may use swaps to take a view on future interest rates or exchange rates.

D. Asset-liability management

Banks and financial institutions use swaps to match the characteristics of their assets and liabilities.

TYPES OF SWAPS

The major types are:



INTEREST RATE SWAP

Meaning

An **Interest Rate Swap (IRS)** is an agreement between two parties to exchange interest payments based on a specified notional principal.

The most common form is:

Fixed interest rate ↔ Floating interest rate

The principal itself is normally not exchanged.

Simple example

Suppose:

- Notional principal = ₹10 crore
- Company A pays fixed interest = 8%
- Company B pays floating interest = MIBOR + 1%

They enter into an interest rate swap.

Interest Rate Swap

₹10 crore notional

Company A <----------> Company B

Pay Fixed 8%

|

▼

Receive Floating

The actual lender continues to receive the payments according to the original loan agreement. The swap changes the economic nature of the interest exposure.

BASIC MECHANICS OF AN INTEREST RATE SWAP

Assume:

- Notional principal = ₹10 crore
- Swap rate = 8%
- Floating rate = 7%

Suppose Company A is the **fixed-rate payer**.

Fixed payment

$$₹10\ crore \times 8\% = ₹80\ lakh$$

Floating payment

$$₹10\ crore \times 7\% = ₹70\ lakh$$

Net payment

$$₹80\ lakh - ₹70\ lakh = ₹10\ lakh$$

Therefore, the fixed-rate payer pays a **net ₹10 lakh**.

Instead of exchanging ₹80 lakh and ₹70 lakh separately, swaps generally use **net settlement**.

NET SETTLEMENT

One of the important features of an interest rate swap is **net settlement**.

Suppose:

- Fixed payment = ₹80 lakh
- Floating payment = ₹70 lakh

Instead of:

- A paying ₹80 lakh
- B paying ₹70 lakh

only the difference is normally paid.

$$₹80 - ₹70 = ₹10\ lakh$$

Therefore:

Fixed-rate payer → pays ₹10 lakh net

CURRENCY SWAP

Meaning

A **Currency Swap** is a derivative agreement in which two parties exchange cash flows denominated in different currencies.

Unlike an ordinary interest rate swap, a currency swap may involve:

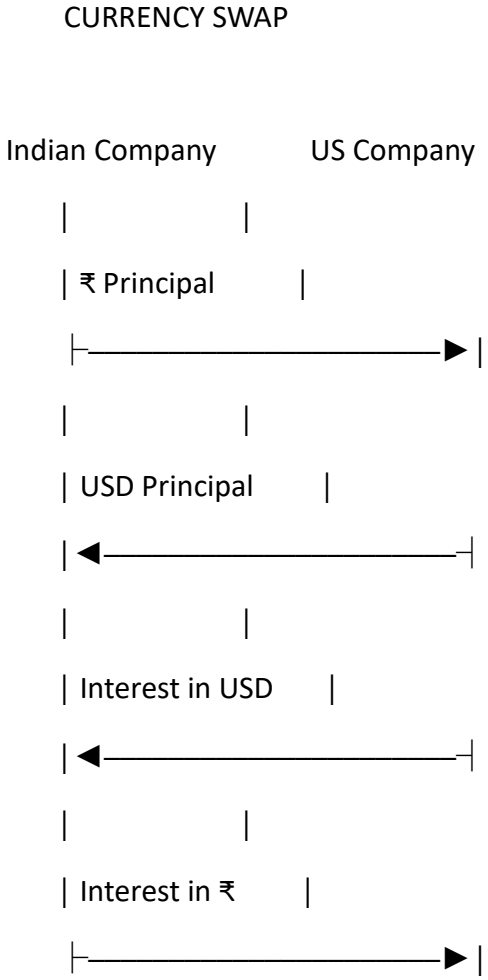
- Exchange of principal at the beginning
- Exchange of interest payments
- Re-exchange of principal at maturity

SIMPLE CURRENCY SWAP ILLUSTRATION

Suppose:

- Indian Company needs US dollars.
- US Company needs Indian rupees.

They can enter into a currency swap.



The exact structure depends on the swap agreement.

EXAMPLE OF CURRENCY SWAP

Suppose:

Indian Company A

Needs:

USD 10 million

US Company B

Needs:

₹83 crore

Assume the exchange rate is:

$$\$1 = ₹83$$

Therefore:

$$\$10\ million \times ₹83 = ₹83\ crore$$

The two companies can exchange principal amounts.

Initial exchange

Company A gives:

₹83 crore

Company B gives:

USD 10 million

During the swap period, interest payments are exchanged.

At maturity, the original principal amounts may be exchanged back.

WHY USE A CURRENCY SWAP?

Currency swaps can be used to:

1. Obtain foreign currency funding

A company can obtain financing in another currency.

2. Manage exchange-rate risk

A company receiving foreign-currency revenues can hedge its exposure.

3. Reduce borrowing cost

A company may have a comparative advantage in its domestic borrowing market.

4. Match cash flows

If a company earns USD but has INR obligations, currency swaps can help align its cash flows.

CURRENCY SWAP – PRACTICAL EXAMPLE

An Indian company has a project in the USA.

It needs:

USD 20 million for 5 years.

The company earns its revenues in dollars, but it normally has easier access to INR financing.

Instead of borrowing directly in USD, it can:

1. Borrow INR.
2. Enter into a currency swap.
3. Exchange INR cash flows for USD cash flows.
4. Use USD for the project.
5. Make the required swap payments.

This can reduce currency mismatch.

INTERST RATE SWAP VS CURRENCY SWAP

Basis	**Interest Rate Swap**	**Currency Swap**
Main purpose	Manage interest-rate risk	Manage currency risk
Currency	Usually same currency	Different currencies
Principal	Usually not exchanged	May be exchanged
Interest	Fixed/floating interest exchanged	Interest in different currencies exchanged
Major risk	Interest-rate risk	Currency + interest-rate risk
Example	Fixed ↔ MIBOR	INR ↔ USD

SHORT NUMERICAL PRACTICE PROBLEMS

Problem 1

A swap has a notional principal of ₹20 crore. The fixed rate is 8% and the floating rate is 9%.

Find the difference in annual interest payment.

$$₹20 crore \times (9\% - 8\%)$$
$$= \boxed{₹20\ lakh}$$

The floating-rate amount exceeds the fixed-rate amount by ₹20 lakh.

Problem 2

A company receives ₹15 lakh after one year. The discount rate is 10%.

Find the present value.

$$PV = \frac{15{,}00{,}000}{1.10}$$
$$= \boxed{₹13{,}63{,}636}$$

Problem 3

A company needs USD 5 million. The exchange rate is ₹84/USD.

Find the equivalent INR amount.

$$\$5m \times ₹84$$
$$= \boxed{₹42 crore}$$